

A Pedagogical Guide to Transport-Projected Exact Diagonalization for Quantum Spin Ice

Implementation note for the `twist_qsi_demo` repository

July 19, 2026

Abstract

This note explains the transport-projected band protocol at the level needed to reproduce, audit, and extend its current implementation. The central problem is that a small periodic pyrochlore cluster contains ice-preserving virtual paths that close only through a periodic image. On cubic-16, a spurious four-spin process then appears at order $J_{\pm}^2/J_{zz}$, parametrically before the contractible six-spin ring exchange at order $|J_{\pm}|^3/J_{zz}^2$ [1]. The protocol labels each completed virtual path by its transport in the covering lattice and projects the selected low-energy band onto zero transport. The production campaign now uses exact microscopic eigenspaces and a polar pullback on matched $M = 2, 3, 4$ character grids; the $M = 3 \rightarrow 4$ centered operator change is 0.498%. The order-two/order-three row enumeration remains only a pedagogical and topological diagnostic.

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1 What problem is being solved?

The local-axis XXZ model is

$$H = H_0 + V = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} \left(S_i^+ S_j^- + S_i^- S_j^+ \right). \quad (1)$$

At $J_{\pm} = 0$, the low-energy model space P is the spin-ice manifold: every tetrahedron has two spins up and two down. In the thermodynamic lattice, the first off-diagonal process within this manifold is the six-spin ring exchange around a contractible pyrochlore hexagon, with scale

$$g_6 = 12 \frac{|J_{\pm}|^3}{J_{zz}^2}. \quad (2)$$

A periodic finite cluster has an additional possibility. A sequence may return to the same finite-cell spin configuration while transporting spin through a nonzero supercell image. Cubic-16 consequently admits ice-preserving four-loops with scale

$$g_4 = 4 \frac{J_{\pm}^2}{J_{zz}}. \quad (3)$$

This is a finite-topology artifact, not a local bulk plaquette. Because it is lower order than g_6 , ordinary periodic exact diagonalization can put the low-temperature peak at the wrong scale even when the microscopic Hamiltonian itself is correct.

The objective is deliberately specific: remove completed virtual processes with nonzero covering-lattice transport from a chosen low-energy band, while retaining contractible processes and the microscopic high-energy complement. This does not by itself prove convergence to bulk quantum spin ice. Order, character resolution, band isolation, cluster size, thermodynamics, and external QMC remain independent validation gates.

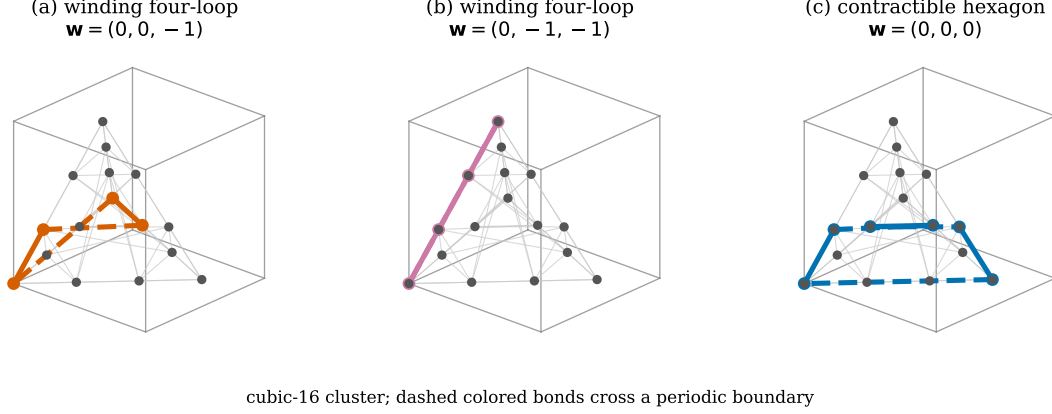


Figure 1: The finite-cluster geometry used by the implementation. The stored bond images distinguish a loop that closes only through a periodic image from a contractible bulk hexagon, even when both look closed after wrapping sites into the simulation cell.

2 Three levels that must not be conflated

Nonperturbative definition. Diagonalize the transported-dipole source family $H(\boldsymbol{\theta})$, identify its isolated ice-descended band, pull every source point into one ice basis with the polar/Kato map, and take a controlled $M \rightarrow \infty$ character limit.

Current production calculation. The active campaign extracts the exact 90-state cubic-16 band in the 12,870-dimensional $S^z = 0$ sector at each independent character point. It uses the primitive $\mathbf{q} = 2\boldsymbol{\delta}$ source, matched $M = 2, 3, 4$ grids, and the exact 65,536-level microscopic complement for all-temperature observables.

Not yet demonstrated. The repository has not established full-Hamiltonian FCC-32 thermodynamics, cluster convergence, an unrestricted gap across all conserved magnetization sectors, or matched QMC-SAC dynamics. The converged cubic-16 heat and entropy curves still fail their QMC tolerances.

This distinction is the most important reading rule for both the manuscript and the code. The formal construction explains what a converged method would mean. The low-order rows explain the mechanism; they do not generate the production spectrum.

3 Geometry and basis construction

3.1 Periodic covering data

Each site has a home-cell position $\mathbf{r}_i$, and the rows of L are the three supercell vectors. Each nearest-neighbor bond stores an integer image $\mathbf{n}_{ij}$. Its displacement in the covering lattice is therefore

$$\mathbf{d}_{ij} = \mathbf{r}_j - \mathbf{r}_i - \mathbf{n}_{ij}^T L. \quad (4)$$

The spin bit pattern alone is insufficient: it forgets which periodic image a virtual hop traversed. The bond image must be propagated through every intermediate state.

The geometry builder performs the following steps:

1. generate wrapped site positions and supercell vectors;
2. find nearest-neighbor bonds and their image vectors;
3. enumerate tetrahedra from the adjacency graph;
4. enumerate bit states satisfying two-in/two-out on every tetrahedron;
5. enumerate ice-preserving simple four- and six-cycles; and
6. lift each cycle to classify its winding.

The resulting census is

cluster	ice states	winding 4	contractible 6	wrapping 6
cubic-16	90	36	16	48
FCC-32	2970	24	32	96

Neither cluster contains a contractible four-cycle.

3.2 Bit basis and excitation energy

A spin configuration is stored as an unsigned integer. Bit $i = 1$ denotes $S_i^z = +1/2$. For a tetrahedron t , let n_t be the number of set bits. After subtracting the common ice energy and setting $J_{zz} = 1$, the relative Ising energy used in denominators is

$$E_0(s) = \frac{1}{2} \sum_t (n_t - 2)^2. \quad (5)$$

Thus every ice state has zero relative energy, while a state with spin-ice charges has positive energy. The recursive ice-state enumerator prunes a partial bit assignment as soon as a tetrahedron can no longer reach exactly two up spins.

4 A virtual path is a data object

The core numerical idea is not a matrix manipulation after the fact. It is a path-resolved construction in which every active row carries

$$(s_{\text{current}}, a_{\text{source}}, \mathbf{N}, c). \quad (6)$$

Here s_{current} is the current microscopic bit state, a_{source} labels the initial ice basis state, $\mathbf{N} \in \mathbb{Z}^3$ is the accumulated bond-image transport, and c is the perturbative amplitude including all denominators accumulated so far.

For every antiparallel nearest-neighbor pair, the perturbation flips both bits. The code contributes a factor -1 for the operator in Eq. (1); the power of $J_{\pm}$ is inserted only when the final row table is

assembled. Depending on whether the active term is $S_i^+ S_j^-$ or $S_i^- S_j^+$, the stored image is updated by $+\mathbf{n}_{ij}$ or $-\mathbf{n}_{ij}$.

Rows may interfere, so duplicates are summed. The merge key is

$$(a_{\text{source}}, b_{\text{target}}, N_x, N_y, N_z), \quad (7)$$

not merely $(a_{\text{source}}, b_{\text{target}})$. Merging without $\mathbf{N}$ would irreversibly combine contractible and winding contributions before they can be projected.

5 What the order-three routine computes

Let $Q = 1 - P$ and define the reduced resolvent, with the ice energy set to zero,

$$R = -QH_0^{-1}Q. \quad (8)$$

Because $PVP = 0$ for this model, the active row routine constructs

$$H_P^{(2)} = PVRVP, \quad (9)$$

$$H_P^{(3)} = PVRV RVP. \quad (10)$$

In code, division by `-e1` or `-e2` applies one factor of R . A path that has already returned to the ice manifold after two hops is recorded in $H_P^{(2)}$ and is not passed through a zero denominator into the third-order table.

The following pseudocode mirrors `sw_order23`:

```
for source, ice_state in enumerate(ice_basis):
    paths1 = apply_V([(ice_state, source, N=0, amplitude=1)])
    paths1.amplitude /= -ising_energy(paths1.state)

    paths2 = apply_V(paths1)
    returned = ising_energy(paths2.state) == 0
    H2_rows += project_to_ice(paths2[returned])

    charged2 = paths2[~returned]
    charged2.amplitude /= -ising_energy(charged2.state)
    paths3 = apply_V(charged2)
    H3_rows += project_to_ice(paths3[ising_energy(paths3.state) == 0])

H2_rows = merge_by_source_target_and_image(H2_rows)
H3_rows = merge_by_source_target_and_image(H3_rows)
```

This routine is a direct implementation of Eq. (10); it is not an arbitrary-order Schrieffer–Wolff engine. The all-order protocol instead fixes the canonical, off-diagonal anti-Hermitian generator

$$S^{(N)} = \sum_{n=1}^N \lambda^n S_n, \quad PS_nP = QS_nQ = 0, \quad (11)$$

so that $Qe^{-S^{(N)}}He^{S^{(N)}}P = O(\lambda^{N+1})$. At each order the off-diagonal Baker–Campbell–Hausdorff coefficient B_n gives

$$S_n = -\mathcal{L}_0^{-1}(B_n), \quad \mathcal{L}_0(X) = [H_0, X]. \quad (12)$$

Implementing Eq. (12) at order four is the next order gate [2].

6 Transported dipole and the zero-transport selector

For a completed row $a \rightarrow b$, let “raised” be sites that are down in a and up in b , and “lowered” the reverse. The implementation computes

$$\delta_{ba} = \sum_{i \in \text{raised}} \mathbf{r}_i - \sum_{j \in \text{lowered}} \mathbf{r}_j + \mathbf{N}^T \mathbf{L}. \quad (13)$$

Coordinates and supercell vectors are multiplied by four and rounded to integers before this calculation. This is exact for the pyrochlore coordinates used here and avoids floating-point equality tests. The factor of four changes units, not the condition $\delta = 0$.

For the exact character family, all completed ice-to-ice paths have even components in these quarter-cell units. The active code therefore uses the primitive integer source $\mathbf{q} = \delta_4/2 = 2\delta$. Using $\mathbf{q} = \delta_4$ would incorrectly alias the leading winding row on the $M = 2$ grid.

Changing a site’s home-cell representative shifts the first two terms in Eq. (13) and $\mathbf{N}^T \mathbf{L}$ oppositely. Hence the transported dipole and, in particular, its vanishing are invariant under this representation choice.

The active assembler has two modes:

$$H_{\text{bare}}^{(3)} = J_{\pm}^2 H_P^{(2)} + J_{\pm}^3 H_P^{(3)}, \quad (14)$$

$$H_{\text{clean}}^{(3)} = \sum_{\delta=0} \left(J_{\pm}^2 h^{(2)}(\delta) + J_{\pm}^3 h^{(3)}(\delta) \right). \quad (15)$$

It then explicitly Hermitian-symmetrizes the dense ice-space matrix. The topology test verifies that the selector removes every winding four-loop and retains every contractible hexagon on both clusters.

7 Why this is a character projection

Attach a phase $e^{i\theta \cdot \Delta \mathbf{q}}$ to each lifted hop. Completed rows have a finite Fourier expansion

$$[H_P^{(N)}(\boldsymbol{\theta})]_{ba} = \sum_{\mathbf{q}} h_{ba}^{(N)}(\mathbf{q}) e^{i\boldsymbol{\theta} \cdot \mathbf{q}}. \quad (16)$$

Average over the characters of $\mathbb{Z}_M^3$:

$$\mathcal{P}_M[H] = \frac{1}{M^3} \sum_{m_x, m_y, m_z=0}^{M-1} H\left(\frac{2\pi}{M} \mathbf{m}\right). \quad (17)$$

Discrete Fourier orthogonality gives

$$\mathcal{P}_M[H_P^{(N)}] = \sum_{\mathbf{q}=0 \pmod{M}} h^{(N)}(\mathbf{q}). \quad (18)$$

At fixed perturbative order only finitely many transports are possible. An M larger than every possible component therefore makes Eq. (17) exactly equivalent to the direct $\delta = 0$ row selector used by the active low-order code. At all orders, however, finite M aliases $\mathbf{q} = M\mathbf{k}$ with zero transport. A controlled $M \rightarrow 2M$ ladder is required.

This also explains why several tempting operations are not equivalent:

- Averaging $Z(\boldsymbol{\theta}) = \text{Tr } e^{-\beta H(\boldsymbol{\theta})}$ averages a nonlinear thermal quantity, not a Hamiltonian block.
- Averaging the microscopic Hamiltonian before multiplication projects individual hops, preventing the virtual paths from composing first.
- Deleting entries of the final ice-space matrix after image labels have been merged cannot recover the decomposition in Eq. (18).
- A fitted counterterm can cancel a chosen leading matrix element but does not classify all completed paths at higher order.

8 Exact isolated-band continuation

Suppose the microscopic Hamiltonian has an isolated exact band projector P_λ with $\text{rank}(P_\lambda) = \text{rank}(P)$. The polar map

$$W_\lambda = P_\lambda P (P P_\lambda P)^{-1/2} \quad (19)$$

canonically transports the model-space basis into the exact band. It fixes the otherwise arbitrary unitary gauge among eigenvectors in that band. The exact pulled-back operator is

$$H_{\text{band}} = W_\lambda^\dagger H W_\lambda. \quad (20)$$

If Ψ is a matrix of exact band eigenvectors and $X = P\Psi$, the code orthonormalizes X as

$$q = X(X^\dagger X)^{-1/2}, \quad H_{\text{band}} = q \text{diag}(E_a) q^\dagger. \quad (21)$$

It reports the minimum, mean, and maximum eigenvalues of $X^\dagger X$ and a unitarity residual. A singular projected Gram matrix means the selected band cannot be canonically identified with the ice manifold; this is a failed band-overlap gate, not a numerical nuisance to regularize away.

For a character calculation, Eq. (19) must be applied separately at each $\boldsymbol{\theta}$ before averaging operators. Averaging eigenvalues alone loses matrix character, while averaging eigenvectors without the polar gauge mixes arbitrary basis choices. The utility functions implementing the polar map and operator average are present in `src/qsicampaign/protocol.py` and are called by the active nonperturbative campaign. At $J_\pm/J_{zz} = 0.046$, the zero-source minimum overlap is 0.762; every generic $M = 3, 4$ point has a larger minimum overlap.

9 Full-temperature spectral replacement

A low-energy effective block alone cannot produce all-temperature thermodynamics. The current cubic-16 construction retains the exact microscopic complement. Let the sorted exact spectrum be $E_1 \leq \dots \leq E_{65536}$ and let $\tilde{E}_1, \dots, \tilde{E}_{90}$ be the exact character-projected band spectrum. First shift the clean band so that its mean equals that of the lowest 90 exact levels, and then define

$$\text{spec}(H_{\text{replacement}}) = \{\tilde{E}_a\}_{a=1}^{90} \cup \{E_i\}_{i=91}^{65536}. \quad (22)$$

Trace matching fixes the band's scalar energy relative to the untouched complement; specific heat within an isolated band would not fix this offset.

There is an important qualification. The exact band is overlap-certified within the conserved $S^z = 0$ sector, where the zero-source gap above it is $0.627J_{zz}$. The cached full spectrum gives a smaller unrestricted gap $0.387J_{zz}$ because other magnetization sectors are present. Those sectors do not mix with the ice band, but an unrestricted spectral-gap statement must still report them separately.

For any finite spectrum, the thermodynamics are evaluated after shifting by the ground-state energy to stabilize the Boltzmann weights:

$$Z' = \sum_i e^{-\beta(E_i - E_0)}, \quad (23)$$

$$\langle E \rangle = E_0 + \frac{\sum_i (E_i - E_0) e^{-\beta(E_i - E_0)}}{Z'}, \quad (24)$$

$$C = \frac{\langle (E - E_0)^2 \rangle - \langle E - E_0 \rangle^2}{T^2}, \quad (25)$$

$$S = \ln Z' + \beta \langle E - E_0 \rangle. \quad (26)$$

All extensive quantities are divided by the number of sites. The temperature grid is logarithmic from $10^{-4}J_{zz}$ to $2J_{zz}$ with 1400 points.

10 External verification against Huang et al.

Huang, Deng, Wan, and Meng simulate Eq. (1) at the sign-free point $J_{\pm}/J_{zz} = 0.046$ using continuous-time worm QMC [3]. This is an especially useful reference because its sign and coupling convention match the repository directly.

The heat-capacity and entropy curves are extracted from the original vector paths in Fig. 1(b) of the arXiv source. The extractor records explicit axis calibrations and writes a CSV with estimated digitization uncertainties. The comparison metric interpolates the finite-cluster curve on $\log T$ and then computes an RMSE over the reference points. Peak gates use fixed temperature windows and log-position errors.

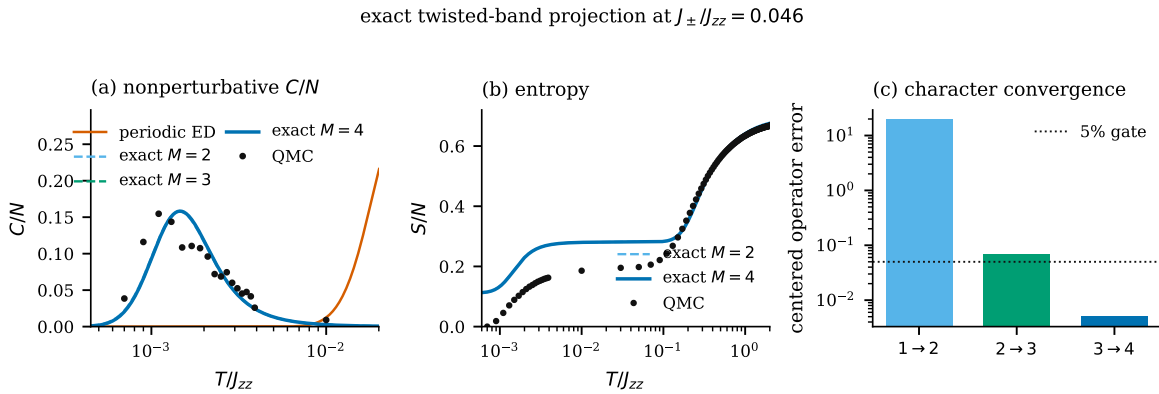


Figure 2: Exact-band benchmark at $J_{\pm}/J_{zz} = 0.046$. The $M = 3, 4$ operators agree, while both converged thermodynamic curves remain distinct from QMC.

The current numerical summary is

quantity	bare cubic-16	clean cubic-16	QMC
low T_p/J_{zz}	0.02554	0.001463	0.001100
low- T C/N RMSE	0.0859	0.02327	0
high T_p/J_{zz}	0.2549	0.2549	0.2105
S/N RMSE	0.0514	0.04509	0

The converged $M = 4$ low-temperature heat-capacity RMSE is reduced by 72.9%, but 0.02327 remains above the threshold 0.02. Its entropy RMSE 0.04509 also fails. Removing winding transfer does not guarantee correct finite-cluster state counting or degeneracy, so these failures diagnose the cluster rather than a lack of character convergence.

The same QMC paper supplies a future dynamical benchmark from stochastic analytic continuation. It reports the sublattice traces

$$S^{zz}(\mathbf{q}, \omega) = \sum_{\alpha=1}^4 S_{\alpha\alpha}^{zz}(\mathbf{q}, \omega), \quad S^{+-}(\mathbf{q}, \omega) = \sum_{\alpha=1}^4 S_{\alpha\alpha}^{+-}(\mathbf{q}, \omega) \quad (27)$$

at $T/J_{zz} = 0.001, 0.04$, and 0.1 . A valid ED comparison must use only momenta in the finite cluster's translation group, match the sublattice trace and spin normalization, use a linear frequency axis, and state its broadening. This dynamics gate remains pending.

11 How the campaign runner fits together

The production data path is short enough to audit directly:

```
T = geomspace(1e-4, 2.0, 1400)

H0, Psi0 = exact_low_band(H(theta=0), dimension=90)
h0 = polar_pullback(H0, Psi0, ice_basis)
h2 = average(exact_gauge_orbit(h0, M=2))

for M in [3, 4]:
    for conjugation representative theta in Z_M^3:
        E, Psi = eigsh(H(theta), lowest=91)
        verify gap, Ritz residual, and ice overlap
        h[theta] = polar_pullback(E[:90], Psi[:, :90], ice_basis)
        h[-theta] = conjugate(h[theta])
    hM = average(h over Z_M^3)

E_full = load cached exact cubic16 spectrum
E_replacement = replace_low_band(E_full, eigvalsh(hM))
compute C/N and S/N from E_full and E_replacement

load Huang vector-extracted C/N and S/N
evaluate fixed gates
write nonperturbative operators, curves, caches, and report
```

No fitting parameter is adjusted in this sequence. The model coupling is fixed to the QMC value and no counterterm or perturbative coefficient enters the production operator.

12 Code map

file or function	responsibility
notes/recompute_finite_size_artifact.py	active geometry, ice basis, cycle census, virtual hopping, order-two/order-three row tables, transported dipole, and dense assembly
build_cluster	positions, bonds, image vectors, tetrahedra, loops, and ice states
apply_v	spin exchange, amplitude sign, and image-transport update
merge_rows	coherent sum at fixed source, target, and image vector
sw_order23	explicit <i>PVRVP</i> and <i>PVRVRVP</i> row construction
transport_delta	integer form of Eq. (13)
assemble	bare or direct zero-transport order-three matrix
src/qsi_campaign/protocol.py	polar pullback, character average, centered operator error, and low-band spectral replacement
src/qsi_campaign/exact_band.py	fixed-magnetization basis, microscopic character Hamiltonian, sparse exact-band extraction, gauge orbit, and overlap/Ritz diagnostics
src/qsi_campaign/thermodynamics.py	stable E/N , C/N , and S/N from a finite spectrum
src/qsi_campaign/benchmarks.py	vector-data loading and log-grid RMSE
campaign/data/extract_huang2018.py	reproducible vector extraction of Huang et al. Fig. 1(b)
campaign/run_nonperturbative.py	active exact $M = 2, 3, 4$ campaign, restartable point cache, thermodynamics, and gate report
campaign/run_validation.py	perturbative topology and cluster-size diagnostic
campaign/make_figures.py	figures generated only from saved campaign products

13 Current gates and their interpretation

gate	status	interpretation
loop topology	pass	winding four-loops removed; contractible hexagons retained
all-temperature stability	pass	high- T peak position unchanged on cubic-16
QMC heat capacity	fail	exact $M = 4$ low- T RMSE $0.02327 > 0.02$
QMC entropy	fail	exact $M = 4$ RMSE $0.04509 > 0.02$
fixed- S^z band gap	pass	minimum source-resolved gap exceeds $0.627J_{zz}$
band overlap	pass	minimum exact-band ice overlap is 0.762
character resolution	pass	centered $M = 3 \rightarrow 4$ error is 0.498%
order convergence	diagnostic only	order 2/3 is not the production method
QMC-SAC dynamics	pending	matched S^{zz} and S^{+-} not computed
FCC-32 full Hamiltonian	pending	current FCC-32 result is ice-manifold order three

All active grids use the same primitive $\mathbf{q} = 2\delta$ source. The large $M = 1 \rightarrow 2$ and 6.78% $M = 2 \rightarrow 3$ changes show why a single parity mask is not enough; the 0.498% $M = 3 \rightarrow 4$ change supplies the

first passed exact-band character gate.

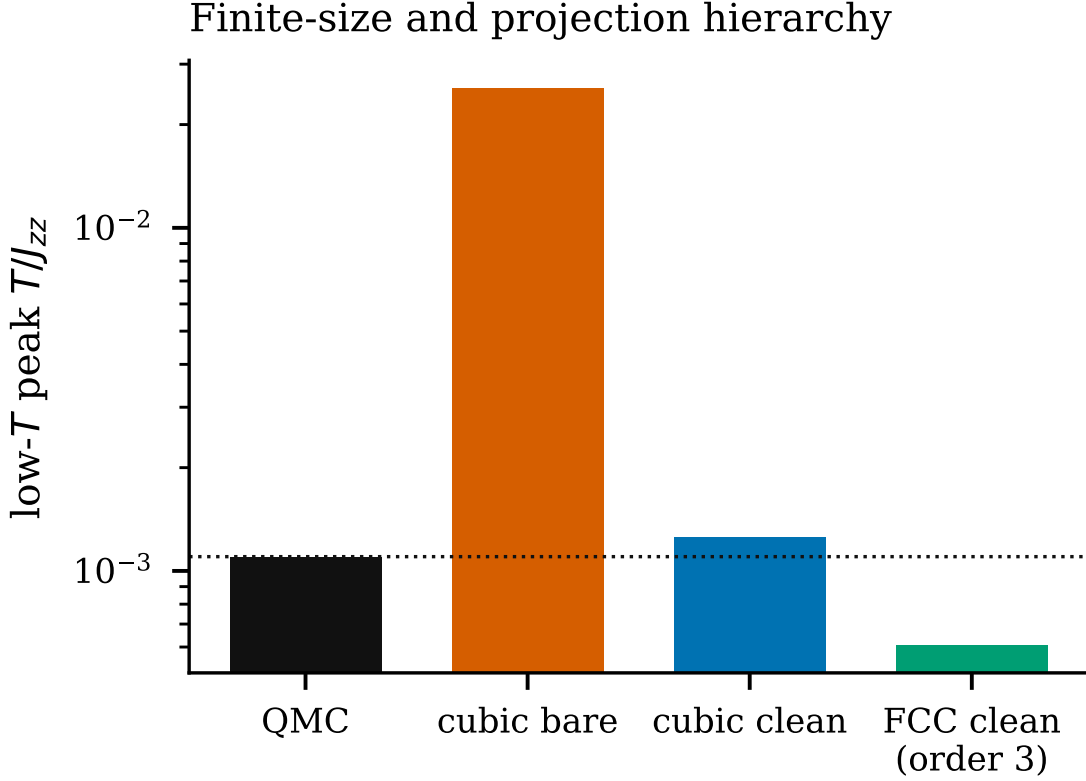


Figure 3: Validation hierarchy. Topological correctness at orders two and three is necessary but not sufficient; the external and convergence gates must be passed before material-level claims are made.

14 Common implementation mistakes

1. **Discarding image transport too early.** Rows must remain keyed by Eq. (7) until after projection.
2. **Calling the order-three block “exact ED.”** Its diagonalization is exact only for that truncated 90- or 2970-dimensional operator.
3. **Calling FCC-32 a full-Hamiltonian result.** The active FCC-32 spectrum contains only the order-three ice block; the 2^{32} complement is absent.
4. **Treating finite M as an exact all-order projection.** Finite characters select $\mathbf{q} = 0 \pmod{M}$, not only $\mathbf{q} = 0$.
5. **Tracking eigenvalues without eigenvectors.** Exact-band identity requires spectral separation and a nonsingular projected Gram matrix.
6. **Averaging thermal observables.** Character projection is linear at the operator level; $e^{-\beta H}$ is nonlinear.

7. **Comparing dynamics at arbitrary momenta.** A finite cluster has a discrete reciprocal translation group. Plot interpolation cannot create missing physical momenta.
8. **Using a heat-capacity peak as the only gate.** The present entropy failure already shows why curve shape and state counting matter.

15 Reproduction and extension checklist

From the repository root, the active workflow is

```
python -m pip install -e .
make all
make nonperturbative-m4
```

It regenerates the row-resolved campaign, runs the unit tests, builds figures, and compiles the manuscript, Supplemental Material, and this note. The machine-readable exact results are `campaign/outputs/nonperturbative_cubic16_p0p046000.npz` and its JSON report. Individual source points are cached under `campaign/cache/nonperturbative_points/`.

An extension should proceed in this order:

1. optionally extend to $M = 5$ as a tighter character check;
2. include all conserved magnetization sectors in the unrestricted gap;
3. diagnose the heat and entropy failures before interpreting the low-temperature peak as bulk convergence;
4. match the QMC-SAC channels at the finite cluster’s actual momenta; and
5. only then deploy a stochastic-complement method on FCC-32 and proceed to a material fit.

16 Takeaway

The protocol’s useful idea is simple: a periodic image label is physical bookkeeping for a virtual path, even though it is invisible in the wrapped spin configuration. Keeping that label until a completed path returns to the ice band makes winding and contractible processes separable. The present implementation proves the path classification at orders two and three and then executes the production projection with exact microscopic eigenspaces. The $M = 3, 4$ agreement removes perturbative-order ambiguity from cubic-16. The remaining heat and entropy failures show that this converged finite cluster still cannot support a bulk or material claim.

References

- [1] Michael Hermele, Matthew P. A. Fisher, and Leon Balents. Pyrochlore photons: The $U(1)$ spin liquid in a $S = 1/2$ three-dimensional frustrated magnet. *Phys. Rev. B*, 69:064404, 2004.

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- [3] Chun-Jiong Huang, Youjin Deng, Yuan Wan, and Zi Yang Meng. Dynamics of topological excitations in a model quantum spin ice. *Phys. Rev. Lett.*, 120:167202, 2018.